In [17]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib
from matplotlib import cm

from sklearn.decomposition import PCA
from sklearn.manifold import MDS
from sklearn.manifold import TSNE

from matplotlib.lines import Line2D


RANDOM_SEED = 239
```

In [2]:

```
data = pd.read_csv("samples_filtered.counts", sep="\t", header=0, index_col=0)
meta = pd.read_csv("metadata.tsv", header=0, index_col=0, sep="\t")
```

In [3]:

```
data = data.div(data.sum(axis=1), axis=0)
data
```

Out[3]:

	ENSG00000000003.16	ENSG00000000005.6	ENSG00000000419.14
Po4S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000093	4.444970e-08	0.000032
Po4S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000089	2.251895e-07	0.000032
Po4S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000086	3.620071e-07	0.000033
Po4S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000095	6.088260e-08	0.000032
FF9S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000072	0.000000e+00	0.000028
FF9S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000069	0.000000e+00	0.000027
FF9S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000089	0.000000e+00	0.000030
FF9S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000091	0.000000e+00	0.000032
BL6S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000087	0.000000e+00	0.000027
BL6S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000082	0.000000e+00	0.000023
BL6S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000075	0.000000e+00	0.000035
BL6S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000081	0.000000e+00	0.000033
Tr5L_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000080	5.042997e-08	0.000038

	ENSG00000000003.16	ENSG00000000005.6	ENSG00000000419.14
Tr5L_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000077	0.000000e+00	0.000045
Tr5L_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000082	4.244399e-08	0.000032
Tr5L_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000090	0.000000e+00	0.000029
P12_1_7_11_22_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000074	0.000000e+00	0.000027
P12_1_7_11_22_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000073	8.288306e-08	0.000032
P12_1_7_11_22_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000076	2.361656e-07	0.000031
P12_1_7_11_22_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000074	6.685679e-08	0.000032
Park14_7_11_22_1_EKDL220017801-1A_HKLKJDSX5_L3	0.000099	4.808049e-08	0.000028
Park14_7_11_22_1_EKDL220017801-1A_HMNK7DSX5_L4	0.000091	8.141520e-08	0.000026
Park14_7_11_22_2_EKDL220017801-1A_HKLKJDSX5_L3	0.000118	0.000000e+00	0.000029
Park14_7_11_22_2_EKDL220017801-1A_HMNK7DSX5_L4	0.000119	8.430337e-08	0.000029
Po4S_DANs_F1_EKDL230000933-1A_HNHG2ALXX_L2	0.000123	1.078320e-06	0.000031
Po4S_DANs_F2_EKDL230000933-1A_HNHG2ALXX_L2	0.000128	6.776910e-07	0.000031
FF9S_DANs_E1_EKDL230000933-1A_HNHG2ALXX_L2	0.000140	2.926986e-08	0.000039
FF9S_DANs_E2_EKDL230000933-1A_HNHG2ALXX_L2	0.000123	2.685975e-08	0.000039
Bl6S_DANs_H1_EKDL230000933-1A_HNHG2ALXX_L2	0.000043	0.000000e+00	0.000040
Bl6S_DANs_H2_EKDL230000933-1A_HNHG2ALXX_L2	0.000041	0.000000e+00	0.000037
P12_1_DANs_D1_EKDL230000933-1A_HNHG2ALXX_L2	0.000060	5.585505e-08	0.000026
P12_1_DANs_D2_EKDL230000933-1A_HNHG2ALXX_L2	0.000059	0.000000e+00	0.000028
Park_14_cl_4_D_B1_EKDL230000933-1A_HNHG2ALXX_L2	0.000083	0.000000e+00	0.000027
Park_14_cl_4_D_B2_EKDL230000933-1A_HNHG2ALXX_L2	0.000090	0.000000e+00	0.000028

34 rows × 58799 columns

In [4]:

```
meta
```

Out[4]:

ID	Tech.rep	Health.PD	Ergicity	Cells.Msc.SPb	Mutation	Lab.name
Po4S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	1	H	DA	SPb	NO	Po4S
Po4S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	1	H	DA	SPb	NO	Po4S
Po4S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	2	H	DA	SPb	NO	Po4S
Po4S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	2	H	DA	SPb	NO	Po4S
FF9S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	1	H	DA	SPb	NO	FF9S
FF9S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	1	H	DA	SPb	NO	FF9S
FF9S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	2	H	DA	SPb	NO	FF9S
FF9S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	2	H	DA	SPb	NO	FF9S
BL6S_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	1	PD	DA	SPb	LRRK2	Bl6S
BL6S_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	1	PD	DA	SPb	LRRK2	Bl6S
BL6S_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	2	PD	DA	SPb	LRRK2	Bl6S
BL6S_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	2	PD	DA	SPb	LRRK2	Bl6S
Tr5L_7_11_2022_1_EKDL220017801-1A_HKLKJDSX5_L3	1	PD	DA	SPb	Parkin	Tr5L
Tr5L_7_11_2022_1_EKDL220017801-1A_HMNK7DSX5_L4	1	PD	DA	SPb	Parkin	Tr5L
Tr5L_7_11_2022_2_EKDL220017801-1A_HKLKJDSX5_L3	2	PD	DA	SPb	Parkin	Tr5L
Tr5L_7_11_2022_2_EKDL220017801-1A_HMNK7DSX5_L4	2	PD	DA	SPb	Parkin	Tr5L
P12_1_7_11_22_1_EKDL220017801-1A_HKLKJDSX5_L3	1	PD	DA	SPb	Parkin	P12-1
P12_1_7_11_22_1_EKDL220017801-1A_HMNK7DSX5_L4	1	PD	DA	SPb	Parkin	P12-1
P12_1_7_11_22_2_EKDL220017801-1A_HKLKJDSX5_L3	2	PD	DA	SPb	Parkin	P12-1
P12_1_7_11_22_2_EKDL220017801-1A_HMNK7DSX5_L4	2	PD	DA	SPb	Parkin	P12-1
Park14_7_11_22_1_EKDL220017801-1A_HKLKJDSX5_L3	1	PD	DA	SPb	Parkin	Park14 cl4

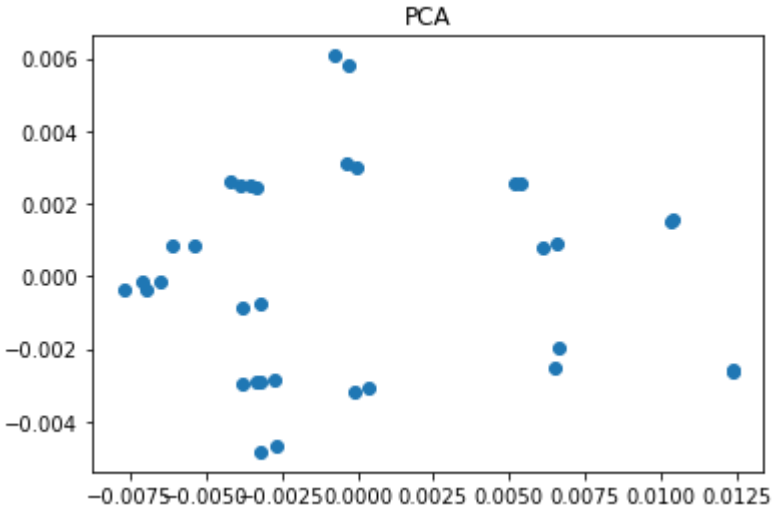
ID	Tech.rep	Health.PD	Ergicity	Cells.Msc.SPb	Mutation	Lab.name
Park14_7_11_22_1_EKDL220017801-1A_HMNK7DSX5_L4	1	PD	DA	SPb	Parkin	Park14 cl4
Park14_7_11_22_2_EKDL220017801-1A_HKLKJDSX5_L3	2	PD	DA	SPb	Parkin	Park14 cl4
Park14_7_11_22_2_EKDL220017801-1A_HMNK7DSX5_L4	2	PD	DA	SPb	Parkin	Park14 cl4
Po4S_DANs_F1_EKDL230000933-1A_HNHG2ALXX_L2	1	H	DA	Msc	NO	Po4S
Po4S_DANs_F2_EKDL230000933-1A_HNHG2ALXX_L2	2	H	DA	Msc	NO	Po4S
FF9S_DANs_E1_EKDL230000933-1A_HNHG2ALXX_L2	1	H	DA	Msc	NO	FF9S
FF9S_DANs_E2_EKDL230000933-1A_HNHG2ALXX_L2	2	H	DA	Msc	NO	FF9S
Bl6S_DANs_H1_EKDL230000933-1A_HNHG2ALXX_L2	1	PD	DA	Msc	LRRK2	Bl6S
Bl6S_DANs_H2_EKDL230000933-1A_HNHG2ALXX_L2	2	PD	DA	Msc	LRRK2	Bl6S
P12_1_DANs_D1_EKDL230000933-1A_HNHG2ALXX_L2	1	PD	DA	Msc	Parkin	P12-1
P12_1_DANs_D2_EKDL230000933-1A_HNHG2ALXX_L2	2	PD	DA	Msc	Parkin	P12-1
Park_14_cl_4_D_B1_EKDL230000933-1A_HNHG2ALXX_L2	1	PD	DA	Msc	Parkin	Park14 cl4
Park_14_cl_4_D_B2_EKDL230000933-1A_HNHG2ALXX_L2	2	PD	DA	Msc	Parkin	Park14 cl4

In [5]:

```
pca = PCA(n_components=2, random_state=RANDOM_SEED)
data_pca = pca.fit_transform(data)
#print(data_pca)
plt.scatter(data_pca[:, 0], data_pca[:, 1])
plt.title("PCA")
```

Out[5]:

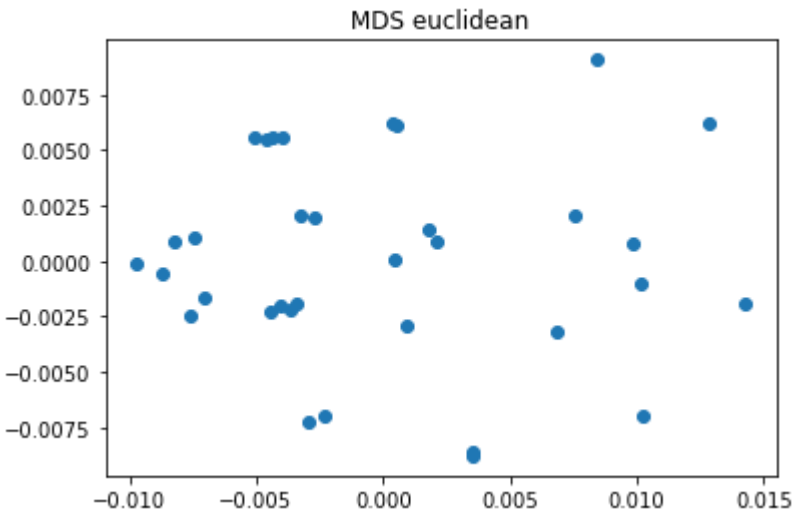
```
Text(0.5, 1.0, 'PCA')
```



In [6]:

```
mds = MDS(n_components=2, metric=True, random_state=RANDOM_SEED)
data_mds = mds.fit_transform(data)
plt.scatter(data_mds[:, 0], data_mds[:, 1])
plt.title("MDS euclidean")
```

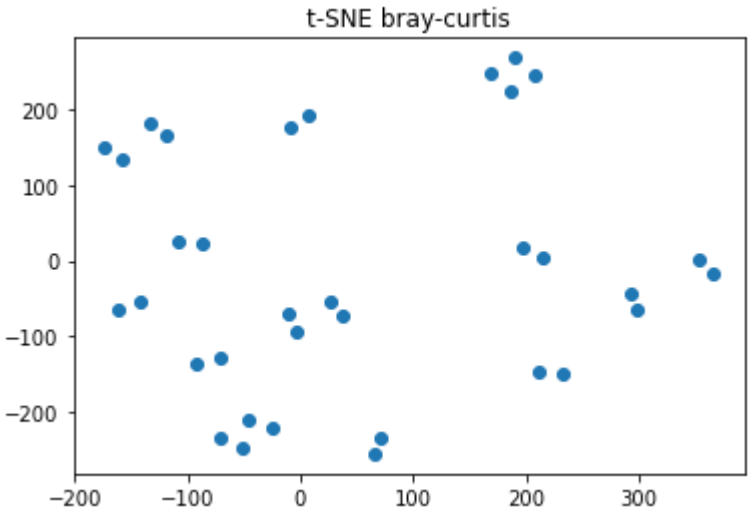
Out[6]: Text(0.5, 1.0, 'MDS euclidean')



In [7]:

```
tsne = TSNE(n_components=2, perplexity=4, metric='braycurtis', random_state=RANDOM_S
data_tsne = tsne.fit_transform(data)
plt.scatter(data_tsne[:, 0], data_tsne[:, 1])
plt.title("t-SNE bray-curtis")
```

Out[7]: Text(0.5, 1.0, 't-SNE bray-curtis')



In [8]:

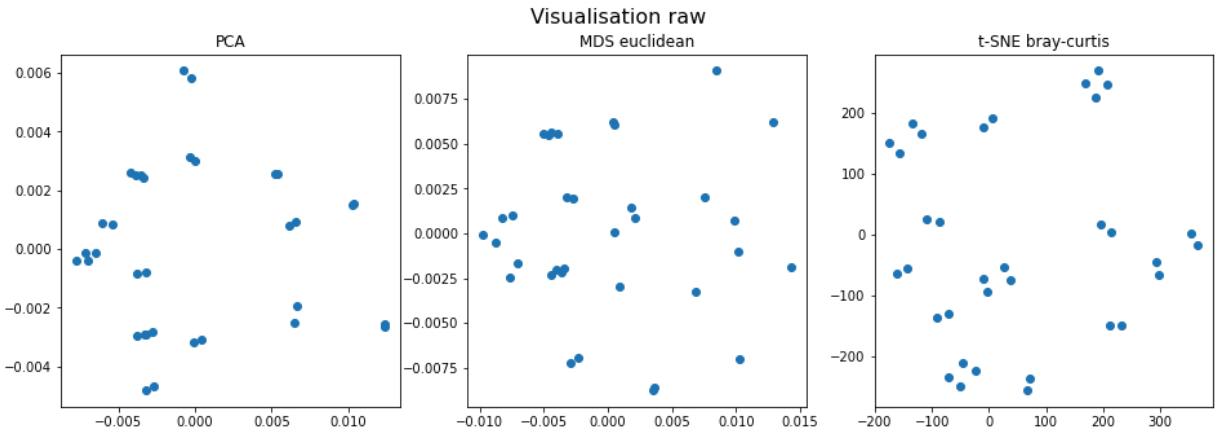
```
fig = plt.figure(figsize=(16, 5))

ax1 = plt.subplot(1, 3, 1)
ax1.scatter(data_pca[:, 0], data_pca[:, 1])
plt.title("PCA")

ax2 = plt.subplot(1, 3, 2)
ax2.scatter(data_mds[:, 0], data_mds[:, 1])
plt.title("MDS euclidean")

ax3 = plt.subplot(1, 3, 3)
ax3.scatter(data_tsne[:, 0], data_tsne[:, 1])
plt.title("t-SNE bray-curtis")

fig.suptitle("Visualisation raw", fontsize=16)
plt.show()
```



In [9]:

```
def get_line(color):
    return Line2D([0], [0], marker='o', markerfacecolor=color, markeredgecolor=color

def draw(params={}, name="Visualisation raw", legend={}):
    fig = plt.figure(figsize=(16, 5))

    ax1 = plt.subplot(1, 3, 1)
    ax1.scatter(data_pca[:, 0], data_pca[:, 1], **params)
    plt.title("PCA")

    ax2 = plt.subplot(1, 3, 2)
    ax2.scatter(data_mds[:, 0], data_mds[:, 1], **params)
    plt.title("MDS euclidean")
```

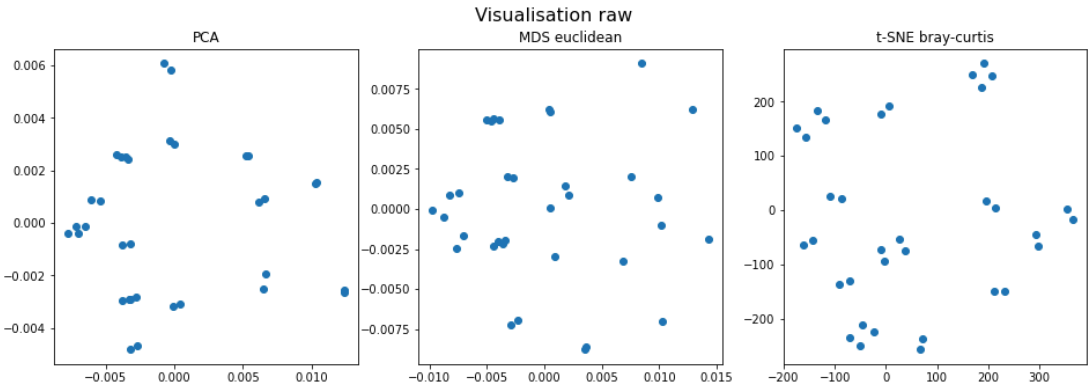
```
    ax3 = plt.subplot(1, 3, 3)
    ax3.scatter(data_tsne[:, 0], data_tsne[:, 1], **params)
    plt.title("t-SNE bray-curtis")


    custom_lines = [get_line(color) for color in legend.values()]
    fig.legend(custom_lines, legend.keys())

    fig.suptitle(name, fontsize=16)
    plt.show()
```

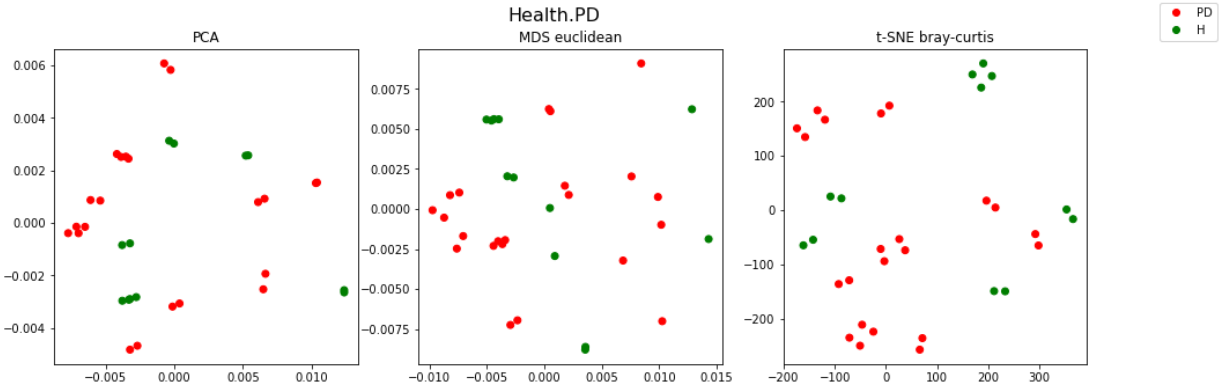
In [10]:

```
draw()
```



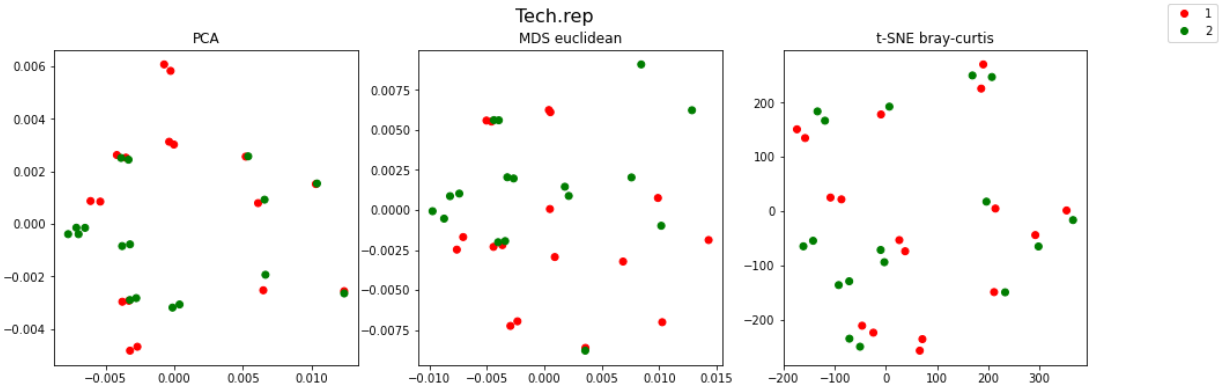
In [11]:

```
# HEALTH.PD
legend = {'PD':'r', 'H':'g'}
params = {'c':[legend[i] for i in meta["Health.PD"]]}
draw(params, "Health.PD", legend)
```



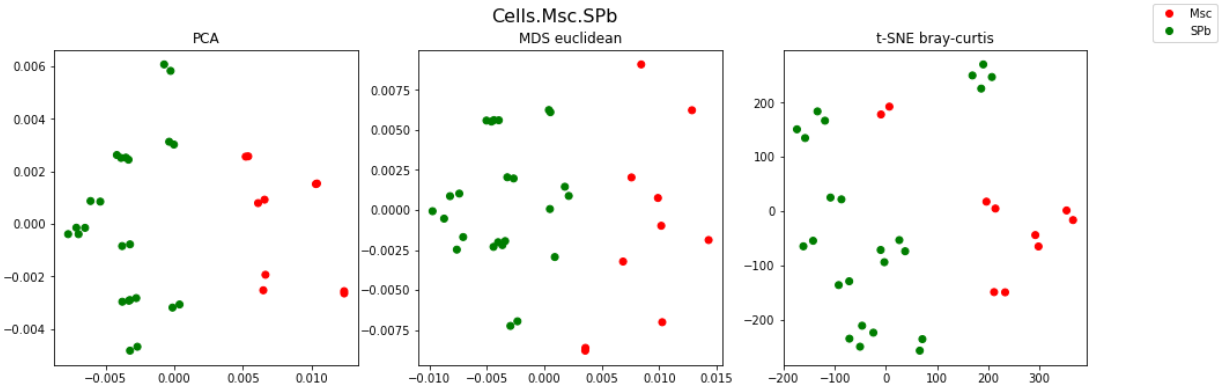
In [12]:

```
# Tech.rep
legend = {1:'r', 2:'g'}
params = {'c':[legend[i] for i in meta["Tech.rep"]]}
draw(params, "Tech.rep", legend)
```



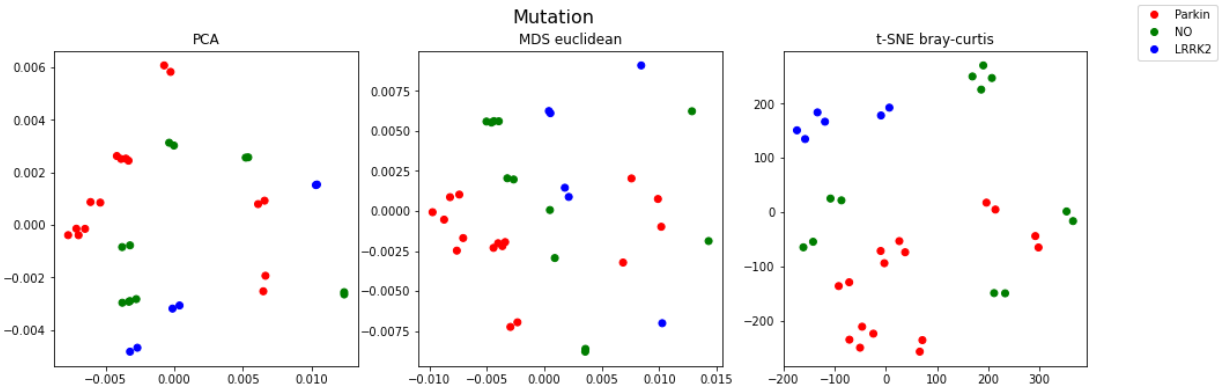
In [13]:

```
# Cells.Msc.SPb
legend = {'Msc':'r', 'SPb':'g'}
params = {'c':[legend[i] for i in meta["Cells.Msc.SPb"]]}
draw(params, "Cells.Msc.SPb", legend)
```



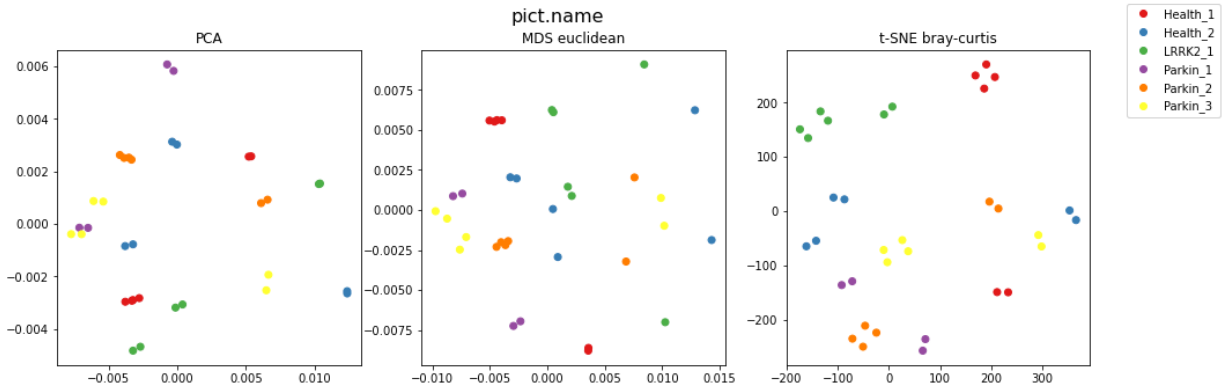
In [14]:

```
# Mutation
legend = {'Parkin':'r', 'NO':'g', 'LRRK2':'b'}
params = {'c':[legend[i] for i in meta["Mutation"]]}
draw(params, "Mutation", legend)
```



In [22]:

```
# pict.name
legend = dict(zip(np.unique(meta["pict.name"]), cm.get_cmap('Set1').colors))
params = {'c':[legend[i] for i in meta["pict.name"]]}
draw(params, "pict.name", legend)
```


pict.name
PCA
MDS euclidean
t-SNE bray-curtis
Health_1
Health_2
LRRK2_1
Parkin_1
Parkin_2
Parkin_3

In [ ]: